



Working with Labeled Volumes

A labeling session generates a groundTruthMedical Object and labeled volume data. These can be viewed and analyzed outside the app.

Navigate to the Module 3 folder in the course files and work through the **AnalyzingLabeledVolumes.mlx** script. You'll work with a completed labeled volume of the MRI data you saw in the previous video to compute properties like volume for each label.

We encourage you to work through the code in MATLAB. A static file is provided below. The code may be different than in MATLAB because interactive tools are not present in static files.

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Note: If this code returns an error, add the course Data folder to your MATLAB Path.

```
segvol = medicalvolume('T1_3T_2018.nii')
```

Most properties, including voxel spacing, number of slices, and plane mapping, are the same as the T1_3T_2018 data you have worked with previously.

The important difference is the values stored in the labeled volume's Voxels property. Use the `unique` function to display the unique values.

```
unique(segvol.Voxels)
```

These values should look familiar. These are the label IDs. The corresponding label names are found in the `LabelDefinitions` table. Background, which is not included in the table, is assigned a label ID of 0 by the Medical Image Labeler App.

Note: Each voxel can only be assigned one label ID.

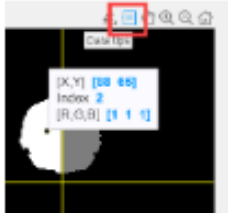
gTruthMed		gTruthMedLabelDefinitions	
gTruthMedLabelDefinitions			
1	2	3	
Name	LabelColor	PixelLabelID	
1 "LHem"	0 0.4470	0.7410	1
2 "RHem"	0.8500 0.3250	0.0980	2
3			

View Labeled Volume

Explore the slices of the labeled volume using the `orthosliceviewer`.

```
orthosliceviewer(segvol)
```

Use the Data Tip tool in the axes toolbar to confirm that voxel values are the label IDs.



Measure Region Properties